Introduction to Data Structures

Course Code: CSC 2106 Course Title: Data Structure (Theory)

Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	**1.1**	**Week No:**	**1**	**Semester:**	
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Lecture Outline

1. Mission, Vision & Goals of AIUB and Its Computer Science Department
2. Course Objectives, Prerequisites, Importance, Contents & Evaluation
3. Classroom Policies
4. Definition of Data Structures
5. Operations on Data Structures
6. Definition of Algorithm
7. Definition of Program
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Vision & Mission of **AIUB**

Vision

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) envisions promoting professionals and excellent leadership catering to the technological progress and development needs of the country.

Mission

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) is committed to provide quality and excellent computer-based academic programs responsive to the emerging challenges of the time. It is dedicated to nurture and produce competent world class professional imbued with strong sense of ethical values ready to face the competitive world of arts, business, science, social science and technology.

Goals of **AIUB**

- Sustain development and progress of the university.
- Continue to upgrade educational services and facilities responsive of the demands for change and needs of the society.
- Inculcate professional culture among management, faculty and personnel in the attainment of the institution's vision, mission and goals.
- Enhance research consciousness in discovering new dimensions for curriculum development and enrichment.
- Implement meaningful and relevant community outreach programs reflective of the available resources and expertise of the university.
- Establish strong networking of programs, sharing of resources and expertise with local and international educational institutions and organizations.
- Accelerate the participation of alumni, students and professionals in the implementation of educational programs and development of projects designed to expand and improve global academic standards.

Vision & Mission of **Computer Science** Department

Vision

Provides leadership in the pursuit of quality and excellent computer education and produce highly skilled and globally competitive IT professionals.

Mission

Committed to educate students to think analytically and communicate effectively; train them to acquire technological, industry and research-oriented accepted skills; keep them abreast of the new trends and progress in the world of information communication technology; and inculcate in them the value of professional ethics.

Goals of **Computer Science** Department

- Enrich the computer education curriculum to suit the needs of the industry- wide standards for both domestic and international markets.
- Equip the faculty and staff with professional, modern technological and research skills.
- Upgrade continuously computer hardware's, facilities and instructional materials to cope with the challenges of the information technology age.
- Initiate and conduct relevant research, software development and outreach services.
- Establish linkage with industry and other IT-based organizations/institutions for sharing of resources and expertise, and better job opportunities for students.

Course Objectives

The objective of this course is to introduce the subject of data structures with the explanation of how data can be stored or manipulated in computer in an optimized way.

An overview of data organization and certain data structures will be covered along with a discussion of the different operations, which are applied to these data structures.

Here, the space and time complexity will be taken care for different searching or sorting techniques to deal with data. We also include how these efficient techniques could be implemented in real life applications.

Course Prerequisites

- Representing information in computers, Binary Number Systems, Conversions.
- Using IDE.
- Basic conception of Data Storage, Data types, Variable, Array (single & multidimensional), Pointers, String, Functions, Recursion, Scope of variable & function, etc.
- Knowing different Libraries & their Functions.
- Concept of Structure & Class.
- Knowing **O**bject **O**riented **P**rogramming concepts.

Importance of the course

- Data structure is required for all areas of computer science – especially for the basic concept of programming.
- This course will give the basic for the understanding of the courses – Algorithms, Database, Artificial Intelligence, object oriented programming, etc.
- This course will give the basic for the understanding of the concepts – Data storage, converting data into information, manipulation of data, etc.

Course Contents

- Mid-term
  - Arrays [1D & 2D]
  - Pointer, String, Structure
  - Stack & Queue
  - Application of Stack & Queue
  - Searching & Sorting
- Final-term
  - Linked Lists [Singly & Doubly]
  - Introduction to Trees
  - Binary Search Tree, Heap Tree
  - Introduction to Graphs
  - Generating **M**inimum **S**panning **T**ree from Graph [Prim's & Kruskal's Algorithms]
  - Graph Traversals [BFS & DFS]

Course Evaluation

Mid-term	2 Quizzes (Best One)	20	**40%**
	Lab Evaluations	20	
	Assignment	10	
	Theory & Lab Class Attendance	10	
	Mid-term Written Exam	40	
	Midterm Total	100	
Final-term	2 Quizzes (Best One)	20	**60%**
	Lab Evaluations	20	
	Assignment	10	
	Theory & Lab Class Attendance	10	
	Final-term Written Exam	40	
	Final Term Total	100	
Grand Total	**Final Grade of the Course**		**100**

Classroom Policies

- To be filled-up by individual course teachers according their requirements…

Data & Structures

- What is *Data*?
  - Data means raw facts or information that can be processed to get results.
- What is *Structure*?
  - Some elementary items constitute a unit and that unit may be considered as a structure.
  - A structure may be treated as a frame where we organize some elementary items in different ways.

Data Structures

Definition

- So, what is *Data Structure*?
  - Data structure is a structure where we organize elementary data items in different ways and there exits structural relationship among the items so that it can be used efficiently.
  - In other words, a data structure is means of structural relationships of elementary data items for storing and retrieving data in computer's memory.

Elements of a Data Structure

- Usually elementary data items are the ***elements*** of a data structure.

- Types of Elementary data items: Character, Integer, Floating point numbers etc.

- However, a ***data structure may be an element of another data structure***. That means a data structure may contain another data structure. For example: Array, Structure, Stack, etc.

- We talk about or study Data Structures in two ways:
    - Basic
        - Having a concrete implementation. Example: Variable, Pointer, Array etc.

    - Abstract Data Types (ADTs):
        - ADTs are entities that are definition of data and operation but do not have any concrete implementation. Example: List, Stack, Queue etc.

Operations on Data Structures

- Basic
  - Insertion (addition of a new element in the data structure)
  - Deletion (removal of the element from the data structure)
  - Traversal (accessing data elements in the data structure)

- Additional:
  - Searching (locating a certain element in the data structure)
  - Sorting (Arranging elements in a data structure in a specified order)
  - Merging (combining elements of two similar data structures)
  - Etc.

Algorithm

Definition

- Set of **instructions** that can be followed to perform **a task**. In other words, **sequence of steps that can be followed to solve a problem**.
- To write an algorithm we do not strictly follow grammar of any particular programming language.
- However its language may be near to a programming language.

Parts of an Algorithm

- Each and every algorithm can be divided into *three sections*:
  - First section is ***input*** section, where we show which data elements are to be given or fed to the algorithm as an input.
  - The second section is the most important one, which is ***operational or processing section***. Here we have to do all necessary operations, such as computation, taking decision, calling other procedures (or algorithms) etc.
  - The third section is ***output***, where we display or get the result with the help of the previous two sections.

Program

- Sequence of **instructions of any programming language** that can be followed to perform **a particular task**.

- Like an algorithm, generally a program has three sections such as **input, processing and output**.

- For a particular problem (usually for a complex problem), at first we may write **an algorithm**. Later, the algorithm may be converted into a **program**.

- In a program usually we use a large amount of data. Most of the cases these data are not elementary items, where exists structural relationship between elementary data items.
    - *That means the program uses **data structures***.

Books

- **“Schaum's Outline of Data Structures with C++”**. By John R. Hubbard
- **“Data Structures and Program Design”,** Robert L. Kruse, 3rd Edition, 1996.
- **“Data structures, algorithms and performance”,** D. Wood, Addison-Wesley, 1993
- **“Advanced Data Structures”,** Peter Brass, Cambridge University Press, 2008
- **“Data Structures and Algorithm Analysis”,** Edition 3.2 (C++ Version), Clifford A. Shaffer, Virginia Tech, Blacksburg, VA 24061 January 2, 2012
- **“C++ Data Structures”,** Nell Dale and David Teague, Jones and Bartlett Publishers, 2001.
- **“Data Structures and Algorithms with Object-Oriented Design Patterns in C++”,** Bruno R. Preiss,

References

1. https://en.wikipedia.org/wiki/Data_structure